

End-to-End Sales Forecasting & Demand Intelligence

Executive Business Report | Prepared for Head of Supply Chain & CFO

1. Executive Summary

We built an intelligent sales forecasting system using 4 years (2015-2018) of Superstore daily transaction data covering multiple product categories and US regions. The system predicts future product demand for the next 3 months, detects unusual sales spikes or drops, segments products by demand behavior, and delivers everything through an interactive dashboard. Our best model (XGBoost) achieves a MAPE of 13.29% after correcting for data leakage, meaning forecasted sales are within 13.29% of actual values on average. We identified 3 distinct product demand clusters with tailored stocking strategies, and detected anomalous weeks driven by seasonal events and promotions.

2. Key Findings from EDA & Forecasting

Revenue by Category: Technology leads with \$827K in total revenue, followed by Furniture (\$729K) and Office Supplies (\$705K). Technology products have the highest average order value.

Regional Performance: West region shows the highest total sales (\$248K in 2018) and the most consistent year-over-year sales growth with stable growth rates. South region shows the highest volatility in annual growth.

Shipping Efficiency: Average shipping time is ~3.96 days across all regions. East region ships fastest (~3.91 days), followed closely by West (~3.93 days), while Central averages the longest at ~4.07 days.

Seasonality: Strong Q4 spikes (October-December) across all 4 years, driven by holiday demand. January-February consistently show the lowest sales. The seasonal decomposition confirms a 12-month cycle.

3. 3-Month Sales Forecast

We built and compared three fundamentally different forecasting models on monthly sales data:

Model	MAE	RMSE	MAPE (%)	Month 1	Month 2	Month 3
SARIMA	\$19,244	\$19,950	20.53%	\$60,332	\$91,458	\$97,168
Prophet	\$20,296	\$22,487	21.89%	\$51,084	\$90,045	\$89,661
XGBoost	\$13,915	\$18,894	13.29%	\$86,466	\$86,507	\$84,327

Recommendation: XGBoost is recommended for production use — it has the lowest MAPE (13.29%) and the lowest MAE (\$13,915), meaning its forecasts are closest to actual sales on average. Forecasted sales for the next 3 months are in the range of \$84K-\$86K per month, showing strong consistency during the peak Q4 sales period.

4. Top 3 Anomalies Detected

Date	Sales	Type	Likely Cause
2015-03-22	\$37,704	Extreme Spike	Possible corporate bulk order or promotional event (Q1 spike)
2018-12-02	\$35,999	Holiday Spike	Festive season / holiday sales rush (early December peak)
2015-02-22	\$225	Deep Slump	Post-holiday demand contraction (lowest recorded weekly sales)

5. Product Demand Segmentation & Stocking Strategy

A KMeans clustering model categorized 17 sub-categories into 3 distinct business segments:

- High Volume, Stable Demand** (Phones, Chairs, Storage, Binders, Accessories, Tables, Machines)
Strategy: Maintain high safety stock, establish automated reorder points to prevent stockouts.

2. Growing Demand (Copiers)

Strategy: Gradually build up inventory, negotiate volume discounts with suppliers to improve margins.

3. Low Volume, Volatile Demand (Art, Fasteners, Paper, Labels, Envelopes, Bookcases, Furnishings, Supplies)

Strategy: Implement Just-In-Time (JIT) replenishment to minimize holding costs, and launch bundle promos to smooth out sales spikes.

6. Actionable Business Recommendations

1. Q4 Supply Chain Preparedness: Because sales surge by over 200% in Q4 (Oct-Dec) compared to Q1, supply chains must preemptively increase stocking levels by late September. Focus safety stock buffers specifically on the **High Volume, Stable Demand** clusters (Phones, Chairs) to capture peak holiday revenues without capital lockup.

2. Capital Allocations for High-Growth Sectors: Allocate marketing budgets and procurement pipelines to support the **Growing Demand** cluster (Copiers), which exhibits an exceptional year-over-year trajectory, while scaling down holding capacity for the **Low Volume, Volatile** group to optimize warehouse efficiency.